



JURUTERA SIBER RAWSEC

Logistics Warehouse Operations Management System

JURUS Level 1 (Analyst) Cyber Engineering Challenge

Role:	Lead Cyber Engineering Analyst	Duration:	Three (3) Weeks
Focus:	Cyber Engineering Infrastructure	Level:	Level 1 Competency

BACKGROUND

A regional logistics and supply chain provider has recently expanded its operations to support multiple warehouse facilities across the country. As part of this expansion, the organisation plans to deploy a centralised Warehouse Operations Management System that enables warehouse personnel to manage shipment tracking, inventory updates, delivery scheduling, operational reporting, and supplier coordination through a web-based platform.

During an internal operational assessment, the organisation identified that its existing IT environment is heavily decentralised, inconsistently configured, and lacks adequate security visibility and operational resilience required to support business-critical logistics services.

The organisation therefore requires a secure and operational infrastructure capable of supporting the new platform while ensuring availability, integrity of operational data, accountability of administrative actions, and recoverability during service disruptions.

Your organisation has been engaged as a cyber engineering consultancy to design and implement the required environment. You have been assigned as the lead Cyber Engineering Analyst responsible for delivering a production-ready solution.

Engineering-First Philosophy

Unlike traditional cybersecurity competitions, no infrastructure, software products, deployment guides, templates, or predefined architectures will be provided. Participants are expected to independently determine the most appropriate technologies, deployment methods, operational controls, security mechanisms, monitoring solutions, and recovery strategies based on practical engineering judgement and industry-relevant practices.

The objective of this challenge is not to evaluate familiarity with any specific technology vendor or platform, but rather to assess the participant's ability to independently design, deploy, secure, monitor, troubleshoot, document, and recover a functional operational environment using sound cyber engineering principles.

COMPETITION DURATION

Participants will be given three (3) weeks to complete the challenge.

During this period, participants may build and deploy their solution using any environment of their choosing.

This may include cloud infrastructure, local virtualisation environments, physical hardware, containerised deployments, or hybrid approaches.

All provisioning, software installation, system configuration, operational testing, troubleshooting, security implementation, documentation, and demonstration preparation activities are the responsibility of the participant.

PROJECT REQUIREMENTS

The logistics provider requires a publicly accessible web-based platform capable of supporting warehouse and shipment management operations.

The chosen solution must be deployed on an operating system selected by the participant and must incorporate appropriate security controls to protect both operational services and stored business data.

The deployed environment must demonstrate secure administrative access controls and appropriate user management practices.

Participants are expected to implement measures that reduce the likelihood of unauthorised access while maintaining operational usability for warehouse administrators and authorised personnel.

The solution should demonstrate evidence of secure baseline configuration practices and appropriate system hardening measures for production-facing infrastructure.

The application environment must include a database component capable of storing inventory records, shipment details, operational logs, and user account information.

Participants are expected to implement suitable database protection mechanisms and demonstrate an understanding of least-privilege access principles, secure credential management, service isolation, and secure configuration practices.

The organisation has also identified a requirement for operational visibility into system and security events.

Participants are therefore required to implement monitoring, logging, or observability capabilities that allow administrators to identify system activity, user actions, operational issues, and potentially security-relevant events.

The technologies selected to fulfil these requirements are entirely at the discretion of the participant.

In addition, the organisation requires assurance that critical services and operational data can be restored following accidental deletion, hardware failure, service interruption, ransomware simulation, or other operational incidents.

Participants must therefore design and implement an appropriate backup and recovery approach capable of restoring business-critical functionality and data integrity.

Participants are encouraged to prioritise maintainability, operational simplicity, service reliability, and practical recovery procedures throughout their implementation.

DELIVERABLES

At the conclusion of the challenge, participants shall submit a technical package containing sufficient evidence to demonstrate the design, implementation, security, operation, monitoring, and recoverability of their solution.

The submission should include:

- Solution architecture diagram
- Technical implementation documentation
- Security configuration evidence
- Administrative access control documentation
- Monitoring and logging evidence
- Backup and recovery procedures
- Operational validation or testing evidence
- Supporting artefacts relevant to the solution

Evidence may include screenshots, configuration files, logs, command outputs, deployment notes, validation results, scripts, diagrams, or any additional material considered relevant by the participant.

Participants should assume that assessors have not previously seen their environment.

Documentation should therefore be sufficiently detailed to allow assessors to understand the participant's design decisions, implementation approach, operational considerations, and applied security controls.

LIVE DEMONSTRATION

Each participant or team will be allocated thirty (30) minutes for a live demonstration and technical defence session.

During this session, participants will be expected to present their architecture and explain the rationale behind their technology selections, deployment decisions, security controls, monitoring implementation, and recovery strategy.

Participants must demonstrate that their solution is operational and capable of supporting the stated business requirements.

Assessors may request demonstrations involving:

- Administrative access management
- Security control implementation
- Database protection mechanisms
- Monitoring and alerting capabilities
- Logging visibility
- Backup validation
- Recovery procedures
- Operational troubleshooting

The demonstration should also provide evidence that participants understand the engineering decisions they have made.

Assessors may ask participants to discuss alternative approaches, explain trade-offs, justify technology selections, and describe how operational and security risks were mitigated.

Participants should be prepared to defend both their technical implementation and the reasoning behind their engineering decisions.

ASSESSMENT PHILOSOPHY

The JURUS programme adopts an engineering-first assessment philosophy.

Assessment will focus on demonstrated practical capability rather than theoretical knowledge alone.

Participants will be evaluated based on:

- Quality and completeness of implementation
- Effectiveness of security controls
- Operational maintainability and usability
- Clarity and completeness of documentation
- Visibility provided through monitoring and logging

- Practicality of backup and recovery procedures
- Ability to explain and justify engineering decisions

A more complex solution will not necessarily receive a higher score. Greater emphasis will be placed on solutions that are appropriate, secure, maintainable, operationally practical, well-documented, and clearly understood by the participant.

The highest-scoring submissions will demonstrate not only technical competence, but also strong engineering judgement, structured operational thinking, and professional communication skills.

EXPECTED OUTCOME

Upon completion of the challenge, successful participants should demonstrate the capability to independently design, build, secure, monitor, troubleshoot, document, and recover a production-ready operational system using industry-relevant cyber engineering practices.

This outcome aligns directly with the JURUS Level 1 Analyst competency objective of developing practitioners capable of independently building and securing systems while applying structured troubleshooting methodologies, operational discipline, and sound engineering judgement.